

CONTACT

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EDUCATION

Vilnius University
Master's degree in Computer Modeling
2019 – 2021

Vilnius University
Additional studies
2018 – 2019

Vilniaus Kolegija
Bachelor's degree, Information systems engineer
2014 – 2017

LANGUAGE

Lithuanian / English / Russian

Igor Vilghelm

Full stack developer

ABOUT ME

Senior Full-Stack Developer with 10+ years of experience building large-scale web systems for energy metering and data processing.

Experienced in system architecture, data visualization, cloud deployment, and CI/CD automation. Led development of complex SCADA-like interfaces and backend systems working with high-volume relational databases and complex data models.

Strong focus on performance, clean architecture, and scalable systems.

EXPERIENCE

Full stack developer / Team lead

2016 – Present

UAB Sigma Telas

Team Lead & Full Stack Developer (2024 – Present)

- Assign tasks to team members and manage workload distribution
- Review code and deliverables to ensure quality and consistency
- Support team members in solving technical challenges and blockers
- Coordinate with cross-functional teams (e.g., DevOps, DB) when needed
- Monitor progress and ensure timely delivery of features and tasks
- Participate in technical decision-making and architecture discussions
- Conducted post-audit security assessments and vulnerability scans using OWASP ZAP, identifying and remediating critical backend security flaws to ensure regulatory compliance
- Improve team processes and development workflows
- Actively contribute to the codebase (backend & frontend development)

Full Stack Developer (2016 – 2024)

- Worked in multiple projects across different domains and business areas
- Designed and developed applications from scratch, including architecture and core systems
- Delivered end-to-end features from concept to production
- Improved application performance and reliability through optimizations, bug fixing, and refactoring
- Implemented features based on business and client requirements
- Maintained and improved existing codebase
- Built and integrated backend services and APIs
- Developed responsive and user-friendly frontend interfaces
- Set up and maintained CI/CD pipelines to streamline development and deployment
- Ensured code quality through testing, code reviews, and best practices

SKILLS

- **Core & Frontend**
  - JavaScript (ES6+), TypeScript
  - React, React Native (Expo), Redux
  - SVG-based SCADA Visualizations
  - Web Security Hardening (CSP, CORS)
  - HTML5 / CSS3 / Tailwind
- **Backend & Architecture**
  - Node.js (Fastify), PHP (Phalcon 5)
  - REST API Architecture & Microservices
  - Redis / Garnet (In-Memory Stores)
  - High-Load Systems & Real-Time Data
  - LLM & AI Integration (Ollama, LM Studio)
- **Databases & Processing**
  - Oracle, MariaDB, PostgreSQL
  - Complex Query Optimization & Tuning
  - Stored Procedures & Functions
  - Time-Series Data Modeling

SCALE & IMPACT

10+ Years — Enterprise Experience
Architecting, developing & maintaining high-load web systems. In IT since 2016.

1M+ — Metering Devices
Real-time automated data processing and monitoring platform.

Per-Second — Aggregation Scale
Time-series data engine from per-second to monthly resolutions.

Multi-Region — Deployments
National-scale systems across Lithuania, Latvia & Central Asia.

PROJECTS

AI Assistant Backend Service for Web Application

2026 – Present

Node.js | Fastify | Ollama | LM Studio | LLM | REST API

Backend service powering an intelligent assistant for the Web Energy Management System with local/hosted LLM integration.

- Developed a backend service using Node.js (Fastify) to power an AI assistant for the Web Energy Management System, enabling users to analyze system events and receive actionable recommendations via a dedicated 'Ask AI' interface.
- Integrated LLM-based solutions (Ollama / LM Studio) to provide contextual explanations of system behavior and assist in decision-making.
- Designed scalable APIs and integrated AI capabilities into the existing system architecture.

Mobile Energy Management System

2025 – Present

React Native | Expo | TypeScript | Offline-First | Mobile UX | REST API

Field engineer cross-platform mobile solution for real-time monitoring and offline data access.

- Designed and developed a mobile energy management application using React Native (Expo) for field engineers, enabling real-time monitoring and data access on the go.
- Focused on performance optimization, offline-first capabilities, and seamless integration with existing backend services.

Web Energy Management System – Maintenance & Evolution

2020 – Present

High-Load | Oracle | MariaDB | Relational DBs | Performance Tuning

Ensuring high availability, resilience, and horizontal scaling of 24/7 mission-critical energy management infrastructure.

- Maintained and evolved a high-load energy management system, ensuring scalability, performance, and reliability in production environments.
- Worked with high-volume relational databases and complex data models.

Web Energy Management System – Architecture & Development

2017 – Present

React | Redux | SCADA SVG | REST APIs | Redis | Garnet

Flagship smart metering & automated commercial energy accounting platform with real-time SCADA scheme viewer.

- Led the design and development of a smart metering system for automated commercial energy accounting.
- Designed and implemented system architecture (frontend & backend), including reusable component libraries and scalable module structure.
- Built a custom component system using React and Redux, enabling consistent UI development across the application.
- Implemented robust Content Security Policy (CSP) rules and hardened IIS security headers to secure frontend assets against client-side injection and cross-site attacks.
- Integrated Redis / Garnet for login rate-limiting and server-side session invalidation, safeguarding databases from brute-force lockouts and enforcing global token revocation.
- Conducted security testing and vulnerability remediation with OWASP ZAP post-cybersecurity audit, resolving critical backend security flaws.
- Developed and maintained backend endpoints for communication with Oracle and MariaDB databases and data acquisition systems (REST APIs).
- Implemented real-time data visualization, including SCADA-like SVG-based interactive scheme viewer & editor with live data binding.
- Worked with large-scale data processing, including Big Data and IoT integrations, data analysis, and data visualization pipelines.
- Optimized application performance and query execution, including analysis and tuning of complex database queries, procedures, and functions.
- Integrated and adapted third-party npm components (charts, grids) to meet specific business requirements.
- Set up and maintained CI/CD pipelines using Jenkins, including custom automation workflows.
- Contributed to cloud adaptation, SSO integration, and infrastructure improvements.
- Provided technical guidance to team members and participated in architecture decisions.

Legacy Web Application Support & Migration

2016 – 2018

ASP 2.0 | Flash ActionScript 2 | Refactoring | New features | Performance | Legacy Migration

Support and progressive modernization of legacy web systems ensuring business continuity.

- Maintained and supported a legacy web application, ensuring system stability and resolving production issues.
- Worked on bug fixing, performance improvements, and gradual refactoring of legacy codebase.

DEVOPS & CLOUD

- Jenkins CI/CD Automation
- Nginx / Apache / IIS
- Cloud Computing & Adaptation
- SSO (Single Sign-On) Integration
- Git Workflows
- Production Monitoring & Troubleshooting

LEADERSHIP

- Team Leadership & Workload Distribution
- Code Reviews & Standards Enforcement
- Technical Mentorship & Blocker Resolution
- Cross-Functional Team Coordination
- Architecture & Technical Decision-Making
- Agile / Workflow Process Optimization

SECURITY & AUDIT

OWASP ZAP Remediation
Remediated critical backend vulnerabilities post cybersecurity audit.

CSP & Hardening
Strict Content Security Policy & IIS production security headers.

Rate Limiting & Token Revocation
Redis / Garnet distributed brute-force protection and session revocation.

VERIFIED LINKS

- Portfolio: vilghelm.eu
- GitHub: github.com/igoriok1994
- LinkedIn: linkedin.com/in/igor-vilghem

KEY TECHNICAL ACHIEVEMENTS

- **Delivered and** deployed energy management systems across multiple regions, including Lithuania and Latvia at different operational levels, as well as national-scale implementations in Central Asia.
- **Built and** scaled a high-load platform processing data from 1M+ metering devices and millions of data points.
- **Engineered a** time-series data system supporting per-second to monthly aggregation levels.
- **Delivered high-performance** data visualization (grids, charts, SCADA-like UI), ensuring smooth UX with large-scale datasets.
- **Led the** architecture and development of a SCADA-like real-time system for energy data visualization and processing.
- **Built and** scaled a high-load data processing system handling large volumes of metering and IoT data.
- **Designed and** implemented a reusable frontend architecture (React + Redux), improving development consistency and scalability.
- **Optimized database** queries and backend performance for complex Oracle and MariaDB systems.
- **Established CI/CD** pipelines using Jenkins, improving development and deployment workflows.
- **Developed a** mobile application (React Native) with offline-first capabilities for field engineers.
- **Designed and** implemented a backend service for AI-powered data querying within the energy management system.
- **Implemented and** integrated an AI Assistant interface into the enterprise web frontend, providing automated contextual data summaries, event analysis, and operational recommendations.
- **Engineered ML-based** anomaly detection and time-series forecasting for metering archives, paired with automated LLM summaries of detected anomalies.
- **Implemented distributed** rate-limiting and centralized session management using Redis / Garnet, protecting authentication endpoints from brute-force/DDoS attacks and enabling immediate multi-device token revocation.